Near-Degeneracy-Aware OLA Inversions Applied to Kepler Red-Giant Observations

Red giants (RG) are prime targets for probing the dynamics of stellar interiors, as their oscillation modes carry information from both the deep core and the outer envelope. Measuring their internal rotation is essential for understanding angular-momentum transport, one of the major open problems in stellar evolution. In particular, stellar rotation splits modes of the same degree into multiple frequency components, providing the only direct probe of internal rotation profiles. Rotation can also induce coupling between closely spaced modes through near-degeneracy effects (NDE). These effects break the symmetry of rotational multiplets and have been shown to bias standard inversion methods, with systematic errors increasing with both rotation rate and evolutionary stage (i.e., decreasing $\Delta\nu$). We use a new OLA-based rotational inversion implementation that explicitly accounts for near-degeneracy effects, including modes of arbitrary degree. The method has been validated using synthetic data and is applied here to real *Kepler* RG targets exhibiting asymmetric rotational splittings. By paving the way for generalized seismic inversion methods on the RG branch, this methodology will provide critical insights into the physical processes that shape stellar evolution.